

2019-20

DISCRETE MATHEMATICS

CSC - 301

Full Marks : 60

Time : Three Hours

The figures in the margin indicate full marks.

Answer any six questions.

Write the answer of all the subparts
of a question together in order.

1. (a) Which one of the first order predicate calculus statements given below correctly express the following English statement? Tigers and lions attack if they are hungry or threatened.

$$(i) \quad \forall x [(tiger(x) \wedge lion(x)) \rightarrow \{(hungry(x) \vee threatened(x)) \rightarrow attacks(x)\}]$$

$$(ii) \quad \forall x [(tiger(x) \vee lion(x)) \rightarrow \{(hungry(x) \vee threatened(x)) \wedge attacks(x)\}]$$

$$(iii) \quad \forall x [(tiger(x) \vee lion(x)) \rightarrow \{attacks(x) \rightarrow (hungry(x) \vee threatened(x))\}]$$

P.T.O.

$$(iv) \quad \forall x \left[(tiger(x) \vee lion(x)) \rightarrow \{ (hungry(x) \vee threatened(x)) \rightarrow attacks(x) \} \right]$$

2

- (b) Investigate the truth value of each proposition, where $P(x, y) : xy > 25$, x and y are real numbers.

(i) $\forall x \forall y P(x, y)$

(ii) $\forall x \exists y P(x, y)$

(iii) $\forall x \exists x P(x, y)$

(iv) $\exists x \forall y P(x, y)$ 2

- (c) Consider the numbers 1, 2, ... , 1000. Show that among any 501 of them, two numbers exist such that one divides the other one. 3

- (d) For how many integers from 1 through 99999, is the sum of their digits equal to 10. Explain your answer. 3

2. (a) Twenty peoples are seated around a circular table. In how many ways can ten pairs of people engage in handshakes so that no arms cross given that nobody is allowed to shake hands with more than one person at once? Assume that all of them shake their hands with each other using right hand only. You must explain your answer. 5

(3)

- (b) Prove that an arbitrary sequence $(x_1, x_2, x_3, \dots, x_{82})$ of distinct integers contains a monotone subsequence of length 10. 3
- (c) Suppose there are " n " bent lines, each containing a zig as shown in Figure 1. What is the maximum number of regions (Z_n) determined by n such bent lines in a plane? Explain your answer. 2

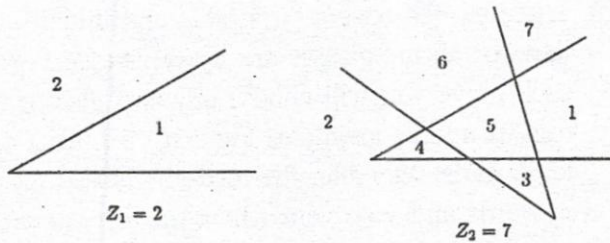


Figure 1: Regions formed by the intersections of two bent lines

3. (a) A 6×6 chessboard can be tiled by eighteen 2×1 dominoes as shown in Figure 2. A tiling is said to be fault-free if any vertical ruling (there are 5 of them) and any horizontal ruling (there are 5 of them) has at least one domino that goes across the ruling. (The tiling in Figure 2 is not fault-free because the horizontal ruling marked with an arrow is not crossed by a domino.) Prove that there is no fault-free tiling of a 6×6 chessboard. 5

P.T.O.

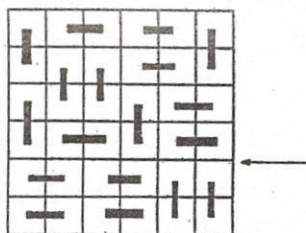


Figure 2 : Faulty tilings of 6×6 chessboard.

- (b) There are 99 glasses on a table and initially the open faces of all the glasses are upwards (see Fig 3(a)). In each move you will choose any two glasses and then turn its upside down (see Fig 3(b)), i.e. if its open face is upwards, then after the move its closed face will be upwards and vice versa. In a move, you may select a glass which was already selected in an earlier move. How many such moves will be required to turn the open face down for all the glasses together? Justify your answer with proper explanation. 5
4. (a) Prove that a tree has either one center or two centers. 2
- (b) The 2^n vertices of a graph G corresponds to all subsets of a set of size n , for $n \geq 10$. Two vertices of G are adjacent if and only if the corresponding sets intersect in exactly two elements. Find out the number of connected components in G . 3
- (c) Suppose that a connected planar graph has six vertices, each of degree four. Into how many regions is the plane divided by a planar representation of this graph? Explain your answer without drawing the graph. 2

(5)

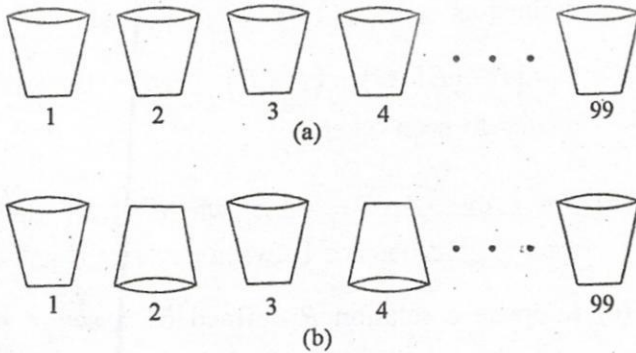


Figure 3: In a move, 2 glasses numbered 2 & 4 are turned upside down

- (d) Let G be a linear graph of 50 vertices. Prove that if the sum of the degrees for each pair of vertices in G is 49 or larger, then there exists a hamiltonian path in G . 3

5. (a) Explain the reason why the set

$A = \{1, 2, 3, 5, 7, 8, 9\}$ under multiplication modulo 10 is not a group. 2

- (b) How many onto functions are there from an n -element ($n \geq 2$) set to a 2-element set? 2

- (c) What is the cardinality of the set of integers X defined as $X = \{n \mid 1 \leq n \leq 123, n \text{ is not divisible by } 2, 3 \text{ or } 5\}$. 3

- (d) If G is an abelian group, then show that

$$(ab)^n = a^n b^n \quad \forall a, b \in G$$

where n is positive integer. 3

P.T.O.

6. (a) Suppose E , F and G be finite sets. The set X and Y defined as $X = (E \cap F) - (F \cap G)$ and

$Y = (E - (E \cap G)) - (E - F)$. How the sets are related to each other. 3

- (b) Prove that $\overline{A \Delta B} = A \Delta \bar{B}$ where $A \Delta B$ defined as symmetric difference between two sets A and B . 3

- (c) Suppose a relation R defined on a set of positive integers such that $\forall x, y \in \mathbb{Z}^+$, $x R y$ iff $x \equiv y \pmod{n}$, ($n \in \mathbb{Z}^+$). Prove that R is an equivalence relation. 4

7. (a) Suppose $f(x) = 2x - 6$ and $g(x) = \sqrt{x}$ be two functions. Determine whether the $g \circ f$ exists or not. If it does not exist, then find a new domain of f for which the composition function $g \circ f$ exists? 3

- (b) Suppose $X = \{a, b, c\}$. Draw the Hasse diagram of $(P(X), \subseteq)$. 4

- (c) Suppose $f: \mathbb{R} \rightarrow \mathbb{R}$ be a function defined by $f(x) = x^2 + x$. Determine whether $f(x)$ is invertible. If so give the inverse function. 3

8. (a) Solve the recurrence relation

$$a_n = 5a_{n-1} - 6a_{n-2} + 8n^2, \text{ where } a_0 = 4 \text{ and } a_1 = 7.$$

(7)

(b) Show that the function $f(n) = 2n^2 + n$ and $g(n) = n^3$ are of the same order. 2

(c) Estimate the growth of the function

$$f(n) = \left| \frac{(3n^4 - 2n)}{(5n - 1)} \right|. \quad 3$$

2019-20

DIGITAL LOGIC DESIGN

CSC - 302

Full Marks : 60

Time : Three Hours

*The figures in the margin indicate full marks.*Answer Question 1 and any *three* from the rest.

15×4=60

1. (a) Convert the following to the other canonical form :

$$F(A, B, C, D) = \sum 0, 2, 6, 11, 13, 14$$

- (b) Design a counter with *T* flipflop that goes through the following binary repeated sequence: 0, 1, 3, 7, 6, 4. Show that when binary states 010 and 101 are considered as don't care conditions, the counter may not work properly. Find a way to correct the design.

5+10=15

2. (a) A sequential circuit has three flip-flops *A*, *B*, *C*; one input *x*; and one output *y*. The state diagram is shown in figure 2. The circuit is to be designed by treating unused state as don't care conditions. Use *D* flip-flops in this design.

P.T.O.

(2)

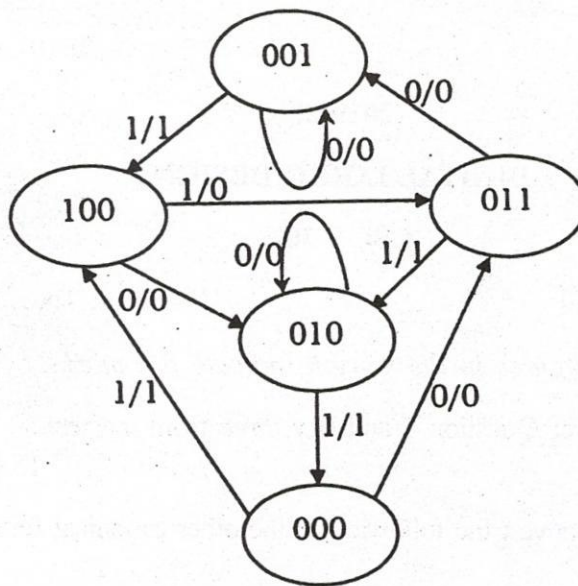


Figure 2

- (b) Simplify the following Boolean function by first finding essential prime implicants :

$$F(A, B, C, D) = \sum(1, 3, 4, 5, 10, 11, 12, 13, 14, 15)$$

and draw the logic diagram for the simplified function.

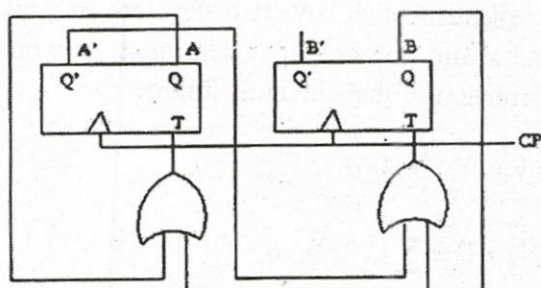
10+5=15

3. (a) Design a four-bit synchronous up-down binary counter using *T* flip-flop.

- (b) Derive the state table and state diagram of the following sequential circuit. What is the function of the circuit?

5+10=15

(3)



4. (a) Show that dual of exclusive-OR is equal to its complement.
- (b) Design a 4-bit register using positive edge triggered D flip flop to operate as indicated below :

Mode Select		Register Operation
A1	A0	
0	0	Hold
0	1	Synchronous Clear
1	0	Complement Content
1	1	Circular Shift Right

5+10=15

5. (a) A combinational circuit is defined by the following three functions :

$$F1 = x'y' + xyz', \quad F2 = x' + y, \quad F3 = xy + x'y'$$

Design the circuit with a decoder and external gate.

P.T.O.

(4)

- (b) A sequential circuit have two flip-flops A1 and A2, one input x and one output y . The next state and output information of that circuit as follows :

$$A1(t+1) = \sum 4, 6$$

$$A2(t+1) = \sum 1, 2, 5, 6$$

$$y(A1, A2, x) = \sum 3, 7$$

Draw the circuit with necessary external gate and its state table.

$$5+10=15$$

Q. No. CSC - 303 / 184

B. Tech./Odd
2019-20/Reg

2019-20

DATA STRUCTURES AND ALGORITHMS

CSC - 303

Full Marks : 60

Time : Three Hours

The figures in the margin indicate full marks.

Attempt any *five* questions.

Write the answers of all the parts of a question in same place with proper sequence.

1. (a) What is the difference between the outputs of the two functions shown below? 3

```
void funl(struct node* head)
{
    if (head == NULL)
        return;

    printf("%d", head->data);
    funl(head->next);
}
```

P.T.O.

```
void fun2(struct node* head)
{
    if(head == NULL)
        return;

    fun2(head->next);
    printf("%d", head->data);
}
```

- (b) Write the algorithms for insertion and deletion in circular queue using array. 4
- (c) Consider the problem of eliminating duplicate elements from a given array of n integers. Design an efficient algorithm to solve this problem. Briefly describe your algorithm with a pseudocode. Also analyze its time complexity. 5
2. (a) Write the Bubble Sort algorithm considering following criteria.
- (i) In a single pass, it will place the highest and lowest elements at the two opposite ends of the array.
 - (ii) It will terminate at any pass once the elements are sorted.
 - (iii) Do not use any for loop. 5
- (b) What do you mean by Radix sort? What is its limitation? 2
- (c) Sort the following keys in descending order using Radix Sort. Clearly show all the steps. Keys are

(3)

759, 67, 15, 354, 91, 879, 125, 43, 79, 83, 245,
543, and 999. 5

3. (a) Briefly describe the difference between array and linked list on the following aspects :

(i) Element access.

(ii) Element insertion and deletion. 4

(b) Describe the concept of adding two polynomials using linked lists. Which type of linked list would you prefer to use for this purpose? 5

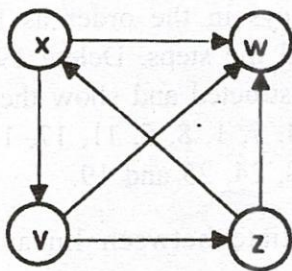
(c) Consider the following pseudocode of a function that takes a number as an argument, and uses a stack S to do processing. What does the following function do? Justify your answer. 3

```
void fun (int n)
{
    Stack S; // Say it creates an empty stack S
    while (n > 0)
    {
        push(&S, n%2); // This line pushes the value of
                        // n%2 to stack S
        n = n/2;
    }
    while (!isEmpty(&S)) // Run while Stack S is not empty
        printf("%d", pop(&S)); // pop an element from S
                                // and print it
}
```

P.T.O.

4. (a) Consider the following scheme for storing binary trees in an array X . Indexing of X starts at 1 instead of 0. The root of the tree is stored at $X[1]$. For a node stored at $X[i]$, the left child, if any, is stored in $X[2i]$ and the right child, if any, in $X[2i+1]$. What should be the minimum size of X such that any binary tree with n vertices can be stored in X ? Justify your answer. 4
- (b) Start with an empty height-balanced binary search (AVL) tree and insert the elements with following keys in the given order: 33, 66, 55, 44, 11, 77, 22, 88, 99. Draw figures to depict your tree after each insertion along with the necessary rotations (if any) for rebalancing. At each step, label all nodes with their balance factors and identify the rotation type (if any). 6
- (c) Use basic properties of red-black tree to prove that the longest simple path from a node x in a red-black tree to a descendent leaf has length at most twice that of the shortest simple path from node x to a descendent leaf. 2
5. (a) Show the adjacency list as well as adjacency matrix representation of the directed graph shown below. Also convert the graph to a strongly connected graph with minimum possible change in direction of the edge(s). 3

(5)



(b) What do you mean by priority queue? Compare the performances of array and heap used for implementation of priority queue. 4

(c) Consider the following array H representing a binary max-heap.

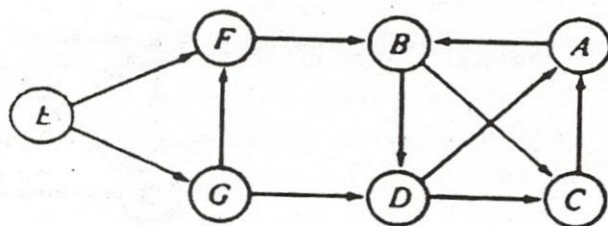
20	17	9	10	11	4	5	8	7	6
----	----	---	----	----	---	---	---	---	---

Suppose that the value of the element $H[1]$ is changed from 17 to 3. Check whether the modified array remains a binary max-heap or not. If not, then make it heap by applying shift-up or shift-down operations as applicable. Show all the steps. Also count the total number of comparisons made in the process of converting the updated version of H into a max-heap. 5

6. (a) Consider a B-tree in which the maximum number of keys in a node is 5. What is the minimum number of keys in any non-root node? Justify your answer. 2

P.T.O.

- (b) Construct a B-Tree of order 5 with the following set of keys in the order as they appear. Clearly depict all the steps. Delete 19 from the final tree thus constructed and show the resultant tree. Keys are 3, 14, 7, 1, 8, 5, 11, 17, 13, 6, 23, 12, 20, 26, 4, 16, 18, 24, 25 and 19. 7
- (c) Differentiate between linear search and binary search. 3
7. (a) Which open addressing technique is free from both primary and secondary clustering? 1
- (b) Clearly show the steps how the elements with keys 18, 54, 22, 44, 46, 58, 31 and 60 are inserted into an initially empty hash table of length 13 using linear probing with hash function $h(k) = k \bmod 13$. 4
- (c) Briefly describe the concept of chaining. Explain how it helps to resolve collision. 3
- (d) Perform the DFS traversal on the graph shown below. Also perform DFS traversal on the corresponding undirected graph. Clearly show the sequence of nodes traversed in each of the two cases. 4



B. Tech End Sem (2019-2020)**Operating Systems****(CSC501)****FM-60; Time 3hrs****Answer any 5 question; each questions carry 12 marks.**

Q1. (a) State Lamport's bakery problem for data access synchronization among a set of 'n' processes sharing a resource 'R'. Can there be a case where multiple process acquire the same token number, if so how it is resolved in the algorithm.

(b) Define a semaphore. Explain the producer-consumer problem (infinite buffer). Provide a solution to the problem using semaphores. Can the problem be solved using just two general semaphores – explain your answer?
((5+1)+(1+4+1))

Q2. (a) Explain the reader-writer problem (weak-reader variation). Provide a solution to the problem using monitor.

(b) State if there is any analogy/difference between the (i) condition variable and semaphore variable, (ii) signal operation in monitor and V operation in semaphore.

(c) State and explain the significance of the different functions used in POSIX semaphore manipulation.
((1+6)+2+3)

Q3. (a) State the limitations of FAT. What are the disadvantages of contiguous allocation.

(b) Provide an outline of the i-node structure. Derive the maximum file size supported in a i-node with 5 direct blocks, 1-single indirect and 1 double indirect. With data blocks size 1MB and assuming each index block can store 10000 indices.

(c) Show in which order the following cylinder requests will be fulfilled considering C-SCAN, FCFS, N-step SCAN, SSTF: 33,60,43,12,120,78,90,32,56 for a disk with 150 cylinders.
(3+(3+2)+4)

Q4.(a) Explain how address translation is carried on in a virtual memory organization that supports both paging-segmentation combined. State the advantage of the above system over pure paging and pure segmentation.

(b) Consider the sequence of page requests : 2,3,5,1,3,7,6,8,3,2,9,8. Calculate the number of page faults corresponding to the following page fault strategies: FCFS, LRU, LFU (least frequently used), Optimal. Comment how LRU can be implemented using clock approximation algorithms.
((5+1)+(4+2))

Q5(a) State the difference between deadlock prevention, avoidance and detection-recovery. State Banker's algorithm and state its purpose.

(b) What are the necessary and sufficient conditions for the occurrence of deadlocks?

(c) What are the necessary conditions that data access synchronization algorithms should satisfy?

(d) Write the pseudo code of the fork() system call.
((1+4)+(1+1+5))

Q6(a) Write a code where the parent process creates 6 child processes using fork() to find the set of prime numbers in a given range from 1 to 'n'. Accept 'n' from user as input. Each child process is supposed find the set of prime numbers in a given range. The overhead of finding should be equally distributed among the children nodes. The children should sync with parent.

(b) Solve the above problem using threads.
(6+6)

2019-20

DATABASE MANAGEMENT SYSTEMS (DBMS)

CSC - 502

Full Marks : 60

Time : Three Hours

The figures in the margin indicate full marks.

Group - A

Answer all questions.

Choose the correct alternative :

2×6

1. Let R_1 (A, B, C) and R_2 (D, E) be two relation schema, where the primary keys are shown underlined and let C be a *f.k* in R_1 referring to R_2 . Suppose there is a no violation of the above referential integrity constraint in the corresponding relation instances r_1 and r_2 . Which one of the following relational algebra expressions would necessarily produce an empty relation?

- (a) $\Pi_D(r_2) - \Pi_C(r_1)$
- (b) $\Pi_C(r_1) - \Pi_D(r_2)$
- (c) $\Pi_D(r_1) \bowtie_{C \neq D} r_2$
- (d) $\Pi_C(r_1) \bowtie_{C=D} r_2$

P.T.O.

(2)

2. SQL views can be used to hide
 - (a) columns and rows only,
 - (b) complicated SQL syntax only,
 - (c) both of the above can be hidden by an SQL view,
 - (d) None of the above.
3. If a denormalization situation exists with a one-to-one binary relationship, which of the following is true?
 - (a) All fields are stored in one relation,
 - (b) All fields are stored in two relations,
 - (c) All fields are stored in three relations,
 - (d) All fields are stored in four relations.
4. A primary key, if combined with a foreign key creates
 - (a) parent child relationship between the tables that connect them,
 - (b) many-to-many relationship between the tables that connect them,
 - (c) network model between the tables that connect them,
 - (d) None of these.
5. In a 1 : N relationship, the foreign key is placed in :
 - (a) either table without specifying parent and child tables,
 - (b) the parent table,
 - (c) the child table,
 - (d) either the parent table or the child table.

(3)

6. The relation book (title, price) contains the title and prices of different books. Assuming that no two books have the same price, what does the following SQL do?

select title from book as B where (select count (*) from book as T where T.price > B.price) < 5

- (a) Titles of the four most expensive books,
- (b) Titles of the fifth most expensive books,
- (c) Titles of the five most expensive books,
- (d) Titles of the fourth most expensive books.

Group - B

Answer any *four* questions.

7. (a) Major airlines companies that provide passenger services in Taiwan are: UniAir, TransAsia Airways, Far Eastern Transport, Great China Airlines etc. Taiwan's Federal Aviation Administration (TFAA) keeps a database with lots of information on all airlines. This information is made accessible to all airlines in Taiwan with the intention of helping the Companies assess their Competitive position in the domestic market. The information kept consists of :

- (i) Each airline has an identification number, name of the contact person and telephone number.
- (ii) For each aircraft identification number, capacity and model is recorded.

P.T.O.

(4)

- (iii) Each employee has an employee identification number, name, address, birthday, sex, position with the company and qualification.
- (iv) Each route has a route identification number, origin, destination, classification (into domestic or international route), distance of the route and price charged per passenger.
- (v) Each airline keeps information about their buy/sell transactions (for example, selling an airplane ticket is a sell transaction, paying for maintenance is a buy transaction)

Draw an E-R diagram for the database presented above. Make sure to identify the associative entity (entities) and provide corresponding key attribute (attributes). 8

- (b) Explain the term "foreign key" with the help of an example and highlight its importance in design of databases. Under what conditions can an attribute of a binary relationship type be migrated to become an attribute of one of the participating entity types? 2+2
8. (a) Show that the axiom of Transitivity is a particular case of the Pseudotransitivity axiom. Use Armstrong's axioms to prove the soundness of the Union rule. 2+2
- (b) Consider the relation Supplier (Supplier-No., Part-No., Supplier-Name, Supplier-Control, Price) and assume that only the following FDs hold for this relation :
Supplier-No $\rightarrow$ Supplier-Name, Supplier-No $\rightarrow$ Supplier-Control. What type of data anomalies does

(5)

this relation have in its present form ? Give examples.
Transform it to 3NF if not already in that form. 2+3

- (c) Consider the set of FDs = {Employee $\rightarrow$ Department, Employee $\rightarrow$ Title, Employee $\rightarrow$ Pay-Scale, Title $\rightarrow$ Pay-Scale}, and find at least two different lossless decompositions for the given set of dependencies. 3

9. (a) Draw initial query tree for the following query and also optimize this query tree. Show each step clearly. 2+4

PJ_{EMPNAME}(EMP JN_{DEPTNUM-DEPTNUM} SL_{MGRNUM=373} DEPT) DF
(SL_{SAL>35000} EMP JN_{DEPTNUM-DEPTNUM} SL_{MGRNUM=373} DEPT)

- (b) Explain why PJNF is a normal form more desirable than is 4NF. 3

- (c) "It is sometimes claimed that every binary relation is necessarily in BCNF". Justify whether this statement is valid or not. Under what conditions can an attribute of a binary relationship type be migrated to become an attribute of one of the participating entity types?

2+1

10. (a) A PARTS file Part# as key field includes records with the following Part# values : 23, 65, 37, 60, 46, 92, 48, 71, 56, 59, 18, 21. Suppose that the search field values are inserted in the given order in a B⁺_tree of order p = 3; show how the tree will expand and what the final tree will look like. Suppose that the following search field values are deleted, in the given order, from this previously constructed B⁺_tree; show how the tree will shrink and show the final tree. The deleted values are : 65, 18, 59. 5+3

P.T.O.

(6)

(b) What are the recovery implications of: 2+2

(i) Forcing buffers to the database at COMMIT?

(ii) Never physically writing buffers to the database prior to the COMMIT?

11. (a) What is recoverable schedule? Why is recoverability of schedules desirable? 2+1

(b) Explain the Two-Phase locking protocol. State the drawback of Two-Phase locking protocol. What is the difference between strict two-phase locking and rigorous two-phase locking? "The timestamp-ordering protocol guarantees serializability" — justify. Write the Thomas write rule. 2+1+2+2+2

12. (a) Consider the table employee (empId, name, department, salary) and the two queries Q1, and Q2 below. Assuming that department 5 has more than one employee, and we want to find the employees who get higher salary than anyone in the department 5, which one of the statements is TRUE for any arbitrary employee table? Justify your answer.

Q1 : Select e.empId

From employee e

Where not exists

(Select * From employee s where
s.department = "5" and s.salary > =
e.salary)

(7)

Q2 : Select e.empId

From employee e

Where e.salary > Any

(Select distinct salary From employee
s Where s.department = "5")

- (i) Q1 is the correct query
 - (ii) Q2 is the correct query
 - (iii) Both Q1 and Q2 produce the same answer.
 - (iv) Neither Q1 nor Q2 is the correct query.
- (b) Consider the following table Cinema (theater, address, capacity)

Which of the following options will be needed at the end of the SQL query

Select P1.address

From Cinema as P1

such that it always finds the addresses of theaters with maximum capacity?

- (i) WHERE P1.capacity > = All (select P2. capacity from Cinema P2)
- (ii) WHERE P1.capacity > = Any (select P2. capacity from Cinema P2)
- (iii) WHERE P1.capacity > All (select max (P2. capacity) from Cinema P2)

P.T.O.

(iv) WHERE P1.capacity > Any (select max (P2.capacity) from Cinema P2)

(c) Consider the following tables in the University database system

department (dept_name, building, budget)

Instructor (ID, name, dept_name, salary)

Section (course_id, sec_id, semester, year, building, room_no, time_slot_id)

- (i) Write an expression in tuple relational calculus to find the names of all instructors whose department is in the Main building.
- (ii) Write an expression in tuple relational calculus to find the Id of the instructors whose salary is greater than Rs. 80,000.
- (iii) Write an expression in domain relational calculus to find the set of all courses taught in the Odd semester of 2018, and in the Even semester of 2019.

(1+2)+ (1+2)+(2×3)

Q. No. CSC 504/

077

B.Tech./Odd
2019-20/Reg

2019-20

MICRO-CONTROLLER BASED SYSTEMS

CSC 504

Full Marks : 60

Time : Three Hours

The figures in the margin indicate full marks.

Answer Question no. 1 and three from the rest questions.

$$3 \times 5 + 15 \times 3 = 60$$

1. Answer the following questions briefly.

- (a) What is the typical characteristic of micro-controller based embedded system?
- (b) What are the real time requirements of the embedded system?
- (c) What are functional requirements of the embedded system?
- (d) Explain the limitation of polling techniques.
- (e) What are the different types of timers in micro-controller based system?

[Turn Over]

2. Design a custom single-purpose processor implementing the GCD of two numbers. Start with a complex state diagram, construct a data-path and a simplified state diagram.
 3. What are the different challenges in designing the micro-controller based system? Explain the types of UART. List the registers associated with UART.
 4. What are the different types of interrupts in 8051? Explain how it is handled by the IE registers. Explain the different addressing mode of 8051?
 5. Can a micro-controller based system, like Aurdino Uno, produce analog output? Explain.
 6. With a neat diagram explain the architecture of the ARM7. What are different operating modes of ARM7. Explain the instruction set of ARM7.
 7. How do you design a power-aware embedded system? Explain properly.
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